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Long distance charging behaviour and range anxiety: An adaptive choice design approach

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Abstract

This paper presents a novel adaptive stated choice experiment formulated as a rolling choice design to capture range anxiety during long-distance travel. It is assumed that respondents have a forward-looking perspective allowing them to select from a set of charging alternatives along the route or postpone charging for a later choice stage. Based on these data, we develop a mixed logit model that reveals several interesting findings. First, we quantify a relationship between the probability to charge and the remaining battery range. Secondly, we find that range anxiety, and thereby battery utilisation between recharges is indeed a heterogeneous quantity that varies among user groups. Tesla drivers and individuals below 50 years of age are less prone to range anxiety compared to other segments. Finally, the results suggest that charging at the early stages of a trip is indeed likely even when the battery level is high.

Keywords: *Charging behaviour, EV range anxiety, Discrete choice, Adaptive choice design, Charging infrastructure, Random utility.*

1 Introduction

Electric vehicles (EVs) are becoming more and more popular, and as a consequence it is important to understand preferences and barriers for the adoption of EVs. Past studies have shown that EV charging infrastructure is important for the roll-out of EVs (and vice versa) (Coffman et al., 2016; Spottle et al. 2018; Springel, 2021, Rich et al., 2022).

One barrier of particular relevance is the fear of running out of battery. This is also known as 'range anxiety' and is considered a determining factor in understanding the charging situation (Rauh et al., 2015). Studies have shown that even though long-distance trips represent a relatively small share of the annual driving needs, this factor is important for EV uptake to facilitate those occasional – but important – trips (Nicholas et al., 2017; Haustein et al. 2021). As pointed out by Mashhoodi and van der Blij (2021), Xu et al. (2020), Bonges et al. (2016) and Guo et al. (2018), range anxiety can have implications for the charging infrastructure design as it influences the charging frequency and battery utilisation more broadly. Specifically, if charging preferences on long-distance trips differ from everyday charging preferences, it is of value to understand how these differ, as the difference could influence the specific design of the charging infrastructure along motorways (Rich et al., 2022). In this regard, several studies highlight the need for more research on consumer preferences related to charging infrastructure, e.g., Hardman et al. (2018), Zhang (2018), Ge (2019), and Visaria et al. (2021).

The range anxiety issue has previously been approached from different angles. However, it involves monitoring EV drivers over longer distances or extended periods to force people to consider charging. More specifically:

- It may be approached from a multiday investigation, where people are asked about charging behaviour over many days until a charging event occurs.
- Or it may be approached from a single-event trip perspective where the trip is long enough to force people to charge.

When users of electric vehicles travel long distances or, more generally, plan their charging scheme, it is a complex mental trade-off between State-of-Charge (SoC) of the battery and the opportunities

offered to the user when considering charging or not (Franke et al., 2012). In principle, both choice situations could be approached from a revealed or stated choice perspective. While revealed choice data have many advantages, it requires a relatively complicated data setup to reflect the specific context. A possible approach is a multiday survey scheme, including monitoring SoC levels over the days. This is challenging due to high dropout rates, non-responses and the fact that such a choice process may need quite a long time horizon, particularly for cars with large batteries and low utilisation. Chakraborty et al. (2019) analyse drivers of charging behaviour based on RP data from EV owners, e.g. location and charging type. However, they omit the specific details regarding the State-of-Charge (SoC), which are essential for understanding the SoC's impact on charging behavior in conjunction with charging infrastructure characteristics.

While the multi-day design represents a more relevant design for many people as it considers everyday routines, it also represents a complex situation as days may have different patterns and trip purposes. Hence, it is likely that range anxiety is masked in other details.

The ‘single-trip’ representation is a simpler framing of the problem. However, while the single-event trip approach might be easier to adapt to a revealed data collection context, it is still challenging. In general, drivers carry out only a few long trips and find difficulty remembering specific things about the trips, such as SoC level, where the charging took place, waiting time, and possible detours. Another strategy would be to monitor the car based on GPS or a dongle. However, this would still introduce the problem of alternative strategies.

A more straightforward data collection strategy is to apply stated choice data. This has been done in some studies to analyse the influence of various attributes on charging decisions, see e.g. Wolbertus and Gerzon (2018), Wen et al. (2016), Visaria et al. (2022), Ge and MacKenzie (2022).

However, to include the influence of SoC, it has to be recognised that the values of various charging locations are interdependent and related in a complicated manner. Hence it is not straightforward how to develop such a choice experiment and how to estimate it.

A few studies have previously used data to investigate the role of SoC in mid-trip decisions on charging, see Sun et al. (2015), Sun et al. (2016), and Ge and MacKenzie (2022). Based on SP data, Ge and MacKenzie (2022) find that SoC plays an important role in charging decisions while charging cost, time and detour time affect charging location choice. Based on RP panel data, Sun et al. (2015, 2016) find that SoC, detour distance and cost influence charging location choice among fast chargers.

In this paper, we take on this challenge and contribute to the literature by i) formulating an adaptive design to investigate charging situations in contexts where range anxiety is integrated into the design, ii) estimating a mixed logit model with a parametrization that allows us to capture the boundary conditions for probability to charge and remaining range (RR), iii) illustrating and discussing how the long-range charging curve differs from everyday charging, and iv) showcasing that range anxiety in this context is indeed heterogeneous. In the next section, we present the novel adaptive choice modelling framework in a long-distance charging context. Section 3 describes the development of the adaptive stated choice experiment and the subsequent data collection. Section 4 presents and discusses the findings from our study, while Section 5 concludes the main take aways from this research.

2 Methodology

2.1 Adaptive stated choice design

In conventional stated choice experiments individuals are presented with a sequence of independent choice experiments also referred to as 'choice tasks' or simply 'tasks'. In adaptive choice designs, choice situations are dynamically changed on the basis of previous choices, attributes or a combination of the two (Fudenberg and Strzalecki, 2015; Eckstein and Wolpin, 1989). As a consequence, the assumption of independence cannot be upheld. Here we describe an approach with an adaptive mechanism that allows us to apply standard estimation procedures and avoid dynamic procedures.

In our formulation we rely on a typical random utility maximisation (RUM) approach (Train, 2009; Ben-Akiva and Lerman, 1985). Let U_{nzti} define the utility function of individual n choosing alternative i in stage t in choice task z. Fundamentally, the utility function can be expressed as

$$U_{nzti} = V_{nzti}(\beta_{zti}, X_{nzti}) + \vartheta_{ni} + \epsilon_{nzti}, \forall n, z, t, i \quad (1)$$

where $V_{nzti}(\beta_{zti}, X_{nzti})$ represents the systematic part of the utility function, ϑ_{ni} is a normal distributed error term that accounts for panel effects for alternatives as well as unobserved correlation among observations for the same respondents, and ϵ_{nzti} are IID extreme value type 1 error terms. This model is known as a mixed logit (ML) model (sometimes referred to as error-component models).

The conditional probability of individual n choosing alternative i in stage t in choice task z is then given by

$$P_{nzt}(i|\beta_{zti}, \vartheta_{ni}) = \frac{exp(V_{nzti}(\beta_{zti}, X_{nzti}) + \vartheta_{ni})}{\sum_{j \in D\hat{}(z,t,j)} exp\left(V_{nztj}\left(\beta_{ztj}, X_{nztj}\right) + \vartheta_{nj}\right)}, \forall n, z, t, i \quad (2)$$

For adaptive surveys (Danaf et al., 2020) the formulation of the underlying choice model relies on the ability to exogenously initialize the system and subsequently describe the menu-generating function. The latter function represents the probability of menu t conditional on the information up to that point, e.g. choices up to menu $t-1$ and all of the corresponding attributes.

$$Q: (X_0, \dots, X_{t-1}, d_0, \dots, d_{t-1}) \rightarrow X_t, \forall n, z, t, i \quad (3)$$

In the present adaptive design, the Q-function is deterministic and choices and attributes for choice stage $t-1$ fully defines the choice-set for stage t. It hereby follows that Q is essentially an indicator function with value 1, if the information up to the point of $t-1$ result in presenting X_t and zero otherwise.

2.2 A 'rolling design' to adapt charging behaviour

The specific choice experiment introduced here resembles a charging situation and is adaptive in the sense that the choice situation is rolled forward in time through a number of charging stages.

At time $t = 1$ the respondent is presented with three charging alternatives $i_1 \in \{1,2,3\}$ and a 'postpone' alternative $i = 0$ that represents the choice of postponing charging to a later time.

At time $t = 2$, the driver is moved forward on his/her artificial route. In doing so, we remove choice $i = 1$ as this opportunity is now passed while adding a new alternative $i = 4$. By monitoring the remaining EV range, while progressing through the different stages, it is possible to reveal trade-offs between charging attributes and range anxiety.

The adaptive nature of the choice set for the different stages is illustrated below in Table 1.

Distance to charging station (km)	$\boldsymbol{i}$	$\boldsymbol{t = 1}$	$\boldsymbol{t = 2}$	$\boldsymbol{t = 3}$	$\boldsymbol{t = 4}$
	0	V_0	V_0	V_0	*Not possible to postpone charging*
10	1	V_1	*Not available – driven by*	*Not available – driven by*	*Not available – driven by*
40	2	V_2	V_2	*Not available – driven by*	*Not available – driven by*
70	3	V_3	V_3	V_3	*Not available – driven by*
100	4	*Not yet considered – too far away*	V_4	V_4	V_4
130	5	*Not yet considered – too far away*	*Not yet considered – too far away*	V_5	V_5
160	6	*Not yet considered – too far away*	*Not yet considered – too far away*	*Not yet considered – too far away*	V_6

Table 1: The rolling stated choice design in a charging context. Initial range in is set to 180±10 km

In the present design, where charging options are unlabelled, we assume that there is only one random effect. The combined dynamic choice experiment is always designed so that charging is required at some point. Hence, the initial remaining range is defined such that the respondent cannot reach the final destination without charging. As a consequence, the final choice does not include a choice alternative of not charging.

2.3 Model specification

In our model specification the activation of the different choice options can be controlled by a non-availability matrix $A = \{a_{it}\}$ where rows i represent choice alternatives from the universal choice set $D = \{0,1, \dots ,6\}$ and t the various choice stages. The matrix A is presented in Eq. (4) below.

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

As seen, the matrix corresponds to Table 1 and includes at all stages an opt-out choice '0' (i.e. postpone charging) and three charging alternatives. The only exception is the final choice situation where the driver is forced to charge to avoid running out of battery.

The choice probability of respondent n choosing a charging option at time $t = 1$ is the probability of choosing an exact option conditional on the probability of not opting out. The choice probability at later stages t' involve the probability of not choosing to charge before arriving to stage t'. In the following, we consider the combined index $\{i, t\} \in I \times T$. If we consider all sequences, we can express the probabilities that a person will choose to charge at time t as in Eq. (5) below.

$$\begin{aligned} &t = 1\text{: } \tilde{Q}_{nz1}(i > 0) = P_{nz1}(i > 0) \\ &t = 2\text{: } \tilde{Q}_{nz2}(i > 0) = P_{nz2}(i > 0)P_{nz1}(i = 0) \\ &t = 3\text{: } \tilde{Q}_{nz3}(i > 0) = P_{nz3}(i > 0)P_{nz1}(i = 0)P_{nz2}(i = 0) \\ &\dots \\ &t = k\text{: } \tilde{Q}_{nzk}(i > 0) = P_{nzk}(i > 0)\prod_{k'=1}^{k-1} P_{nzk'}(i = 0) \end{aligned} \quad (5)$$

Hence, if a person n chooses to charge at stage $t = k$ in choice task z, then $\tilde{Q}_{nzt}(i)$ is composed of the joint probability of not charging up to that point in time multiplied by the probability of charging for $t = k$.

As commented above, the design is always such that there is precisely one charging decision for each adaptive choice sequence. For ease of notation, we define the entries of the availability indicator matrix A in Eq. (4) as in Eq. (6) below.

$$a_{nz}(t,i) = \begin{cases} 1 & \text{alternative } i \text{ is available at time } t \\ 0 & otherwise \end{cases} \quad (6)$$

Given the conditional probabilities in Eq. (2), the combined unconditional probability of selecting charging alternative $i \in D$ over all time periods T, can be stated as in Eq. (7).

$$P_n(\boldsymbol{\beta}) = \int_{\vartheta} \prod_{z=1}^{Z} \prod_{t=1}^{T} \prod_{i \in D} a_{nz}(t,i) \tilde{Q}_{nzt}(i|\boldsymbol{\beta}) f(\vartheta) d\vartheta, \quad \forall z,t,i \quad (7)$$

Here we model panel dependencies by means of a mixed logit approach for the ϑ_{ni} error term. The generic likelihood function can now be stated as in Eq. (8).

$$LL(\boldsymbol{\beta}) = \sum_{n=1}^{N} \ln(P_n(\boldsymbol{\beta})^{y_n}) \quad (8)$$

where y_n is an indicator that indicates (equals one) if the choice sequence of individual n includes alternative i at time t in choice task z and zero otherwise. Due to the integral in Eq. (7) the model is estimated by using maximum simulated likelihood (Train, 2003).

3 Data

3.1 Stated Choice Design

For the generation of the stated choice design, we have applied the software package Ngene (ChoiceMetrics, 2018). Each choice task consists of 7 alternatives (6 charging locations labelled A-F and the opt-out alternative when charging is postponed). The experiment is designed to mimic the dynamic choice behaviour for longer trips and with particular focus on the interplay between remaining range and charging attributes.

In the design, we split each choice task into multiple instances, for which we only present three charging alternatives at a time. That is, in the first 'stage' an initial remaining range is drawn at random (within certain bounds) and the respondent can choose between charging options A, B, and C as well as the

option to postpone charging. If the respondent selects option A, B, or C the choice experiment halts as we only allow one charging to take place during the hypothetical trip. If the respondent chooses not to charge, the design is rolled forward with a certain distance. In the new choice situation faced by the user, alternative A is no longer an option (as this alternative is now behind). However, a new charging station D is now included. The battery level and distance to charging stations are updated for each stage and this process continues until the respondent has chosen a charging option. The stated choice experiment is designed in such a way that charging station F will be the last possible option before the battery runs flat. Thus, when reaching the last 'stage', the choice task alternatives D, E, and F are presented to the respondents, while the 'postpone' alternative is now unavailable. This ultimately forces the respondent to charge in the last 'stage'.

We defined two stated choice designs which differ only in the number of attributes included. The attributes included in the two designs are shown in Table 2. Both of the designs (basic and complex) are efficient designs (zero priors) and include 40 choice tasks divided into 20 blocks. In other words, each respondent is presented with 2 choice tasks from the basic design as well as 2 choice tasks from the complex design.

It is assumed that the expected waiting time is known prior to the charging decision. For the calculation of waiting time, a generic waiting time formula based on queuing theory can be applied (Vandet and Rich, 2023). With respect to charging speed, it should be noted that while charging speed is generally a non-linear function that depends on a variety of factors (such as the battery level and type), it is here assumed that the function is linear. Hence, respondents that charge for 5 minutes will obtain half the added range compared to a 10-minute charging. It was also considered if speed should be presented by referring to the effect of the chargers, e.g., 100 or 200 kW chargers. However, pilots revealed insignificant parameters and it was decided to use this slightly more generic representation which seemed more robust. All attributes are defined according to five levels, except for facilities which have only three levels. Level values are presented in Table 3 below.

Attributes	Description	Basic design	Complex design
Remaining Range	This attribute states the current remaining range of the battery expressed in kilometres. This is a context specific variable that is not related with any of the charging alternatives.	yes	yes
Distance	This variable represents the distance to the charging station (in combination with the remaining range) and is included to quantify range anxiety, i.e. the amount of battery buffer individuals prefer before charging.	yes	yes
Detour	This attribute expresses how much individuals are willing to deviate from their main route, or how strongly they prefer charging options that are located directly along the main road without detour.	yes	yes
Waiting Time	This attribute measures the waiting time at the station before a charger becomes available.	yes	yes
Cost	The cost attribute defines a unit price for 1 kWh. As respondents might not charge a similar amount due to differences in preferences as well as differences in battery size we present cost as a unit price per kWh, rather than a total charging cost. As the sample represents EV users it is assumed that these are familiar with kWh prices.		yes
Charging Speed	This variable quantifies the rate at which a device's battery is replenished during the charging process. Since it is not known how much individuals prefer to charge, we cannot provide a charging time. Instead, we present a charging speed attribute that expresses the added range (in km) they gain per 10 minutes of charging.		yes
Facilities	We include several different measures of facilities at the stations, e.g., toilet, shopping, or restaurant facilities.		yes

Table 2: Attributes use in the different experiments.

Note here that the attribute *Distance* (to the charging station) is included to add some randomness in the distance between charging stations to mimic real life uncertainty more closely. More specifically, by default we assume charging stations are located every 30 km, however the exact distance between each charging station can vary with up to +/- 10 km, as represented by the levels in Table 3.

Figure 1 shows how a specific choice stage is presented to the users. The specific choice stage is taken from the complex design where cost, charging speed and facilities are included (Appendix A includes the subsequent pages in the rolling design for a complete overview). In reality, range uncertainty exists due to differences in weather conditions and driving style. To handle such uncertainty, we introduce an interval around the expected remaining range in the experiment rather than a specific number. More specifically, in the specific example presented in Figure 1 the initial remaining range is defined as 185 in the design, however, to reflect some degree of range uncertainty the remaining range is presented as an interval of ±10 km of the initial remaining range, i.e. between 175-195 km.

Also note that all respondents are told that they must complete a journey of 280 kilometers. The trip length reflects a situation where respondents are forced to charge. Hence, the trip length is well above the initial remaining range of the battery (median range is defined as 180 km as presented in Table 3). On the other hand, to keep the choice context as simple and transparent as possible, we aim to prevent respondents from considering multiple charging instances during their trip. Including such scenarios would complicate both the choices and the stated choice experiment. Thus, we believe that a trip length of an additional 100 kilometers compared to the (median) initial range was appropriate to ensure that the trip was adequate for a single charge.

Ultimately, it is technically feasible to offer all six charging options (A-F) at each stage, including past and future ones. However, for our design, we decided to incorporate only the three closest upcoming charging stations to enhance efficiency and user experience. This was to mimic a choice context in a setting where there would be “sufficient” charging stations (i.e., like the number of gas station), where respondents do not have to meticulously plan their exact charging location in advance, but rather assess the situation as the trip progress. Also, despite it being possible to turn-around a go back to a charging station that is behind (which in some instances might be the closest charger), we consider passed charging stations to be “out of mind”, and thus only consider the three nearest forthcoming charging stations.

		Levels				
Attributes	**Unit**	**1**	**2**	**3**	**4**	**5**
Remaining Range	Km	170	175	180	185	190
Distance	Km	-10	-5	+0	+5	+10
Detour	Min	0	3	6	9	12
Waiting Time	Min	0	5	10	15	20
Cost	DKK/kWh	2	4	6	8	
Charging Speed	km/10 min	60	75	90	105	120
Facilities		Toilet	Toilet & Restaurant	Toilet & Shop		

Table 3: Attribute choice levels.

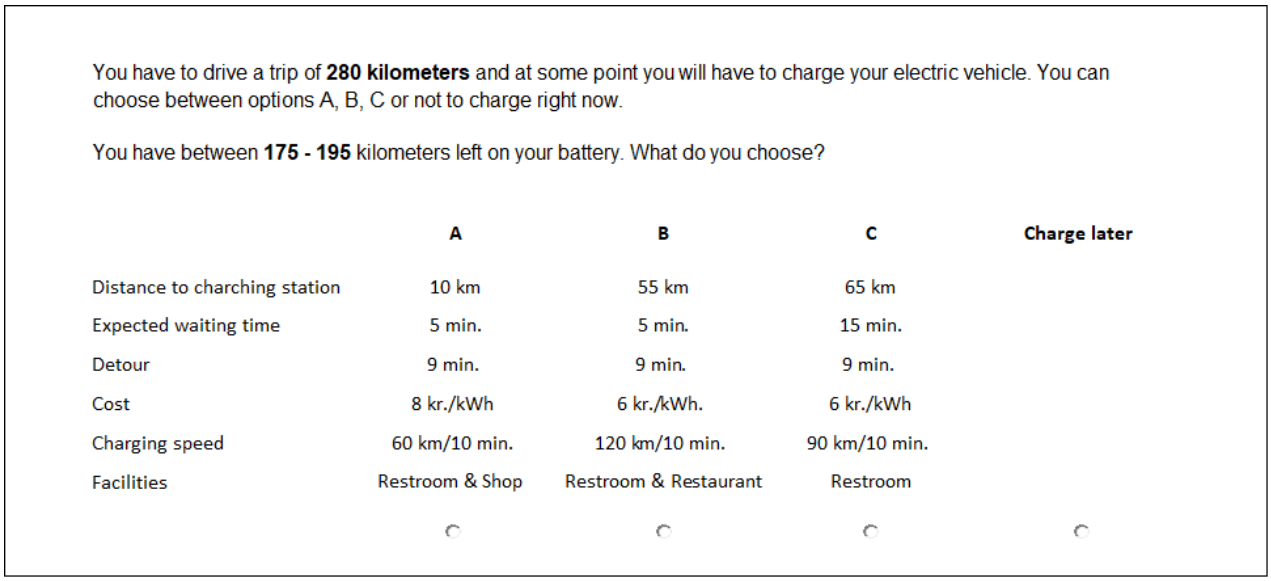


Figure 1: Illustration of the first choice stage in the complex stated preference design.

3.2 Data Sample

The data were collected by sending invitations to the approximately 2,000-3,000 members of the Danish electric car association, *United Danish Electric Motorists* (Forenede Danske Elbilister, FDEL). To avoid GDPR issues, a third-party consultant was hired to handle the email distribution.

455 complete survey responses were obtained. This yielded a response rate of approx. 15-20%. We removed answers from respondents who did not have an electric vehicle, did not answer the stated

choice experiment, or were the recorded data was inconsistent[1]. The final sample consisted of 354 respondents. Each respondent was presented with 2 simple experiments and 2 complex experiments. The choice distribution is presented in Table 4, and it is noted that 42% charge in the very first game, while approx. 80% select the midrange charging options B, C, D, or E.

		A	**B**	**C**	**D**	**E**	**F**	**Total**	***Percent***
Stage	**Remaining range**	145-185 Km	115-155 km	85-125 Km	55-95 km	25-65 km	-5-35 km		
1	170-190 km	70	84	145	0	0	0	**299**	*42.2%*
2	140-160 km	0	17	30	66	0	0	**113**	*16.0%*
3	110-130 km	0	0	8	25	60	0	**93**	*13.1%*
4	80-100 km	0	0	0	75	62	66	**203**	*28.7%*
Total		**70**	**101**	**183**	**166**	**122**	**66**	**708**	*100.0%*
Percent		*9.9%*	*14.3%*	*25.8%*	*23.4%*	*17.2%*	*9.3%*	*100.0%*	

Table 4: Distribution of choice alternatives.

Table 5-Table 8 presents descriptive statistics for various dimensions of the data. It can be noted that the sample is biased towards elderly men with higher education. The majority have a personal parking space in a driveway or garage and have a charging unit installed. Tesla is the dominant car manufacturer, and the average battery size is just below 60 kWh. We emphasize that our sample of EV owners is not reflective of the overall Danish population. It is more closely related to the population of EV owners, which is also skewed toward older and richer individuals. However, a chi-square test of independence still rejects that the sample population is statistically equivalent to the population of EV owners when comparing across gender, age and income. Although this lack of representativeness may limit the generalizability of our results to the national level, it does not preclude us from exploring variations within our sample. Specifically, we can still investigate differences between groups of early and late adopters.

[1] An observation is considered inconsistent if a respondent has declared to charge in stage 4, however one (or more) of the previous stages had not recorded that the postpone option was selected.

		Adaptive design	
		Frequency	Percent
Gender	Female	28	7.9%
	Male	321	90.7%
	Unknown	5	1.4%
Age (years)	18-29	1	0.3%
	30-49	77	21.8%
	50-69	203	57.3%
	70+	70	19.8%
	Unknown	3	0.8%
Household income after tax (DKK/month)	Less than 30,000	64	18.1%
	30,000 - 39,000	61	17.2%
	40,000 - 49,000	70	19.8%
	50,000 - 59,000	59	16.7%
	60,000 - 69,000	35	9.9%
	70,000+	30	8.5%
	Unknown	35	9.9%
Education	Vocational school	47	13.3%
	Short-term further (1½ - 2 years)	31	8.8%
	Medium-term further (2 - 5 years)	120	33.9%
	Long-term further (min. 5 years)	127	35.9%
	Other	26	7.3%
	Unknown	3	0.8%

Table 5: Descriptive statistics on the socio-demographic variables.

		Adaptive design	
		Frequency	Percent
Housing Type	Urban	136	38.4%
	Sub-urban	129	36.4%
	Rural	89	25.1%
Car parking	Public/Street	36	10.2%
	Shared parking space in complex	9	2.5%
	Reserved parking space in complex	5	1.4%
	Personal parking in driveway or garage	304	85.9%
Home Charging Unit	Installed	301	85.0%
	Possible but not installed	15	4.2%
	Not Possible	38	10.7%

Table 6: Statistics related to housing, parking, and home charging.

		Adaptive design	
		Frequency	Percent
Make	BMW	15	4.2%
	Hyundai	70	19.8%
	Nissan	52	14.7%
	Renault	29	8.2%
	Tesla	103	29.1%
	VW	31	8.8%
	Other	54	15.3%
Battery Size (kWh)	Less or equal to 44 kWh	111	31.4%
	Greater than 44 kWh	243	68.6%
Purchase Year	2018-2020	304	85.9%
	2015-2017	40	11.3%
	Before 2015	9	2.5%
	Unknown	1	0.3%
Other Cars in household	More than one BEV	52	14.7%
	Gasoline/Diesel Cars	128	36.2%

Table 7: Statistics related to EVs and other cars in household.

		Adaptive design	
		Frequency	Percent
Long Trips	Yes	182	51.4%
	No	172	48.6%
Problems	Payment solution	47	13.3%
	Connection to the charger	52	14.7%
	Too few chargers	86	24.3%
	Chargers occupied/long waiting time	74	20.9%
	Only/mainly slow chargers	51	14.4%
	Other problems	25	7.1%

Table 8: Statistics related to long trips and relevant problems with charging.

4 Results and discussion

4.1 Econometric specification

One of the challenges of the econometric specification is that it should allow us to model the proper boundary conditions for the probability to charge as a function of the remaining range. This requires that the probability approaches one when remaining range approaches zero. The final utility function that allows us to capture this boundary condition is presented in Eq. (9).

$$
\begin{aligned}
U_{PostponeCharging} &= 0 \\
U_{nzti} &= \alpha_{ASC} + \vartheta_{n1} \cdot \sigma_{ASC} \\
&+ \beta_{\frac{1}{Remaining\ Range}} \cdot \frac{1}{Remaining\ Range_{nzt}} \\
&+ \big(\beta_{Remaining\ Range-Distance} + \sigma_{Remaining\ Range-Distance} \cdot \vartheta_{n2} + \beta_{\text{Age<50}} \\
&\qquad \cdot \text{Dummy}(\text{Age} < 50)_n + \beta_{Tesla\ Owner} \cdot Dummy(Tesla\ Owner)_n\big) \\
&\qquad \cdot (Remaining\ Range_{nzt} - Distance_{nzti}) \\
&+ \big(\beta_{Detour} + \sigma_{Detour} \cdot \vartheta_{n3}\big) * Detour_{nzti} \\
&+ \big(\beta_{\text{Waiting Time}} + \sigma_{\text{Waiting Time}} \cdot \vartheta_{n4}\big) * \text{Waiting Time}_{nzti} + \epsilon_{nzti} \\
&\forall n = 1, \dots, N, t = 1, \dots, T, i \in D
\end{aligned}
\tag{9}
$$

The final utility specification is derived after thorough testing of systematic and random preference heterogeneity, as well as non-linear effects. As can be seen, when $Remaining\ Range_{nzt} \rightarrow 0$ the utility of charging $U_{nzti} \rightarrow \infty$ for $\beta_{\frac{1}{Remaining\ Range}} > 0$. In the adaptive design we found all attributes to have unobserved preference heterogeneity, which is here expressed by the standard normally distributed error terms $\vartheta_{n1}, \dots, \vartheta_{n4}$.

The Mixed Logit (ML) model is estimated with the software package PandasBiogeme v3.2.8 (Bierlaire, 2020) using 10,000 (0,1) NORMAL MLHS draws. The model accounts for panel effect, and thus the draws are sampled per individual (not observation). The complex design turned out, despite a promising prior pilot test, to perform poorly with parameters being insignificant and/or having counterintuitive sign. For example, we found a positive parameter for charging costs[2]. Ultimately, we decided to focus on the simpler – and apparently – more robust basic design, which include remaining battery range, distance to charging station as well as the expected waiting time and detour associated with that charging station. Parameters and model summary are presented in Table 9.

[2] We suspected that this could be due to income effects or some segments in the sample not caring about cost which would ultimately influence a generic cost-parameters, however, we found the cost parameters to be positive for all income groups except one, which were not significant.

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Parameters	Unit	Value	Rob. t-test	Rob. p-value	Percentage of sample with "correct" sign
α_{ASC}		-2.513	-2.664	0.008	
σ_{ASC}		3.763	8.617	0.000	
β_{Detour}	1/min	-0.123	-7.771	0.000	88.3%
σ_{Detour}	1/min	0.103	3.498	0.000	
$\beta_{Waiting\ Time}$	1/min	-0.078	-7.264	0.000	78.6%
$\sigma_{Waiting\ Time}$	1/min	0.098	6.845	0.000	
$\beta_{\frac{1}{Remaining\ Range}}$	km	0.634	4.921	0.000	
$\beta_{Remaining\ Range-Distance}$	1/km	-0.009	-3.619	0.000	76.2%
$\sigma_{Remaining\ Range-Distance}$	1/km	-0.012	-3.964	0.000	
$\beta_{Age<50}$	1/km	-0.009	-2.469	0.014	92.8%
$\beta_{Tesla\ owner}$	1/km	-0.009	-2.623	0.009	92.6%

Model summary	
Estimated parameters:	11
Respondents:	354
Observations:	708
Stages:	1,616
Final log-likelihood:	-1,421.863
Likelihood ratio test (against null-LL)	2012.408 > χ2(α=0.05, df=10): 18.307
Rho-square-bar (against null-LL)	0.410
AIC	2865.727
BIC	2908.289
Number of draws:	10,000

277 **Table 9:** Parameter estimates and model summary of the final model. The column indicating the percentage of
278 sample with the "correct" sign, is due to the normally distributed error terms.

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The results show that all parameters in the final model specification are significant at the 0.01 level. As expected, the parameters for detour and waiting time are negative, which yield lower probability for choosing an alternative if detour and waiting time increase. The remaining range attribute is included in two ways. Firstly, we included the reciprocal value of the remaining range which provides an intuitive charging probability curve, i.e., when the remaining range goes towards zero the charging probability goes towards 100%. Secondly, we included the residual range ($Remaining\ Range_{nzt} - Distance_{nzti}$), i.e., the remaining battery range when arriving at the charging station for alternative i. As expected, we find this parameter to be negative, yielding that the more (residual) range there is, the lower probability for charging.

We tested mixed parameters for all numerical attributes using normal mixing distributions. This indicated significant random coefficients related to detour, waiting time and the residual battery range when arriving at the charging station. As an unavoidable consequence of the normal mixing distribution, a portion will be positive values as indicate in Table 9. We would like to emphasize that the tails with the wrong sign should not be seen as a representation of preferences but merely as a statistical artefact of the distributional assumptions. Note that for the socio-demographic interactions with the residual range, we assume the same standard deviation in the preference heterogeneity across segments, while the mean of the distribution is dependent on the segment, as defined in Eq. (9). Residual range has the lowest proportion with the correct sign; however, the socio-demographic interactions shift the mean value of the distribution so that a higher share of those respondents has the expected sign. More specifically, 42.1% of the sample owns a Tesla and/or are below 50 years of age (9.0% of the sample belongs to both segments).

Further, besides the socio-demographic characteristics included in the final model specification, we also tested gender, income, children, EV experience and urban density. We only found a significant age effect for individuals below 50 years of age, who are more likely to postpone charging. This suggests that individuals below 50 years of age are less risk averse, which is in line with existing literature (Dohmen et al., 2017). We also found Tesla users to be less risk averse. We suspect this could be related to the fact that Tesla are the market leader with respect to driving range and one of the EV brands which has been the longest on the market. Thus, it is not surprising that Tesla users, on average, have less

range anxiety. We also speculated if there could be a hidden element of experience in driving EVs as Tesla were among the first more widely popular EVs on the market, so some Tesla users are likely to have had more time to get used to the driving of EVs and thus more familiar with range uncertainties. However, when testing for an effect related to years of EV ownership, we found no significant impact on the model.

4.2 Range anxiety and charging curves

One of the most debated terms in the literature of charging behaviour is that of 'range anxiety'. While the term is well-understood in describing the fear that a vehicle has insufficient range to reach its destination, no formal mathematical descriptions have been offered so far. In this paper, to showcase that range anxiety is indeed heterogeneous, we compute the probability for charging as a function of the remaining range on the battery. At the same time this also demonstrates general sensitivity of the model. It should be emphasized that the model reflects the properties of the specific SP sample and has not been calibrated in any way. Still, however, it provides some interesting insight, which we will elaborate on below.

Figure 2 illustrates the probability to charge across socio-demographic segments as a function of the remaining battery level, while Figure 3 present the probability to charge for various level-of-service combinations as a function of the remaining battery level. In both figures the blue line represents the base line probability, where detour and expected waiting time are assumed to be zero. As discussed in the previous section, individuals below 50 years of age and Tesla owners are less risk averse, and Figure 2 shows how this translates to decision making with respect to charging. More specifically, it is shown that Tesla owners and/or individuals below 50 years of age generally have a lower charging probability at any given point in time and reflect that these segments generally have a lower range anxiety. Interestingly, we can also see that non-Tesla owners who are below 50 years of age have almost the same likelihood of charging as 50+ Tesla drivers. Similarly, in Figure 3, we can see if a detour is required to reach the charging station and/or there is (expected) waiting time at the charging station, then the probability of individuals deciding to charge decreases, and the respondents are more likely to postpone charging (compared to the baseline) in hope of a future charging station with lower (or no)

detour and waiting time. As the remaining range decrease however, the detour and (expected) waiting time becomes less relevant as recharging becomes more pressing, and thus the charging curves moves closer to the baseline. The trading between charging options with different expected waiting times has also been considered in Vandet and Rich (2023) in a slightly different context. The paper demonstrated how users of electric vehicles can reduce queuing time at chargers, if waiting time information is shared in the charging system and if people use their battery flexibility to circumvent queuing and plan their charging schemes accordingly.

In Hipolito et al. (2022) a distribution for steady-state SoC was estimated. This distribution represents the SoC distribution over a population of EVs on a given random day. When compared to the steady-state distribution, the profile of the identified distribution (e.g., Figure 5-8) at low levels of SoC, are indeed comparable. However, there are also differences in that the slope of the probability curve in our paper presents itself as being steeper towards $SoC \rightarrow 0$. This is not surprising, as the paper by Hipolito et al. estimated a steady-state distribution with respect to everyday use and with home-charging opportunities. It is natural to expect that for longer trips, for which people are forced to charge, we should see people pushing the battery slightly more towards the limits. Keep in mind that people trade waiting time and detours, which are often irrelevant in everyday contexts.

Another takeaway from Figure 2 and Figure 3 is that the tail of the curve is relatively flat when compared to the previously mentioned steady-state distribution. Hence, the probability to charge even for relative high levels of remaining range is significantly different from zero. We believe there are two reasons for this. Firstly, that the specific design forces people to think strategically when planning their charging. Because they trade waiting time and detours in their charging decision, it is indeed possible that it is efficient to charge at an early stage. In fact, for those EV drivers who have used apps such as "Better route planner" or the internal Tesla app, this appears to be very realistic. The second explanation is that we present only levels of remaining range that, with the current battery ranges, only allow us to realistically capture the left part of the curve where the remaining range is low. Hence, the tail of the presented distribution should indeed be interpreted with some care.

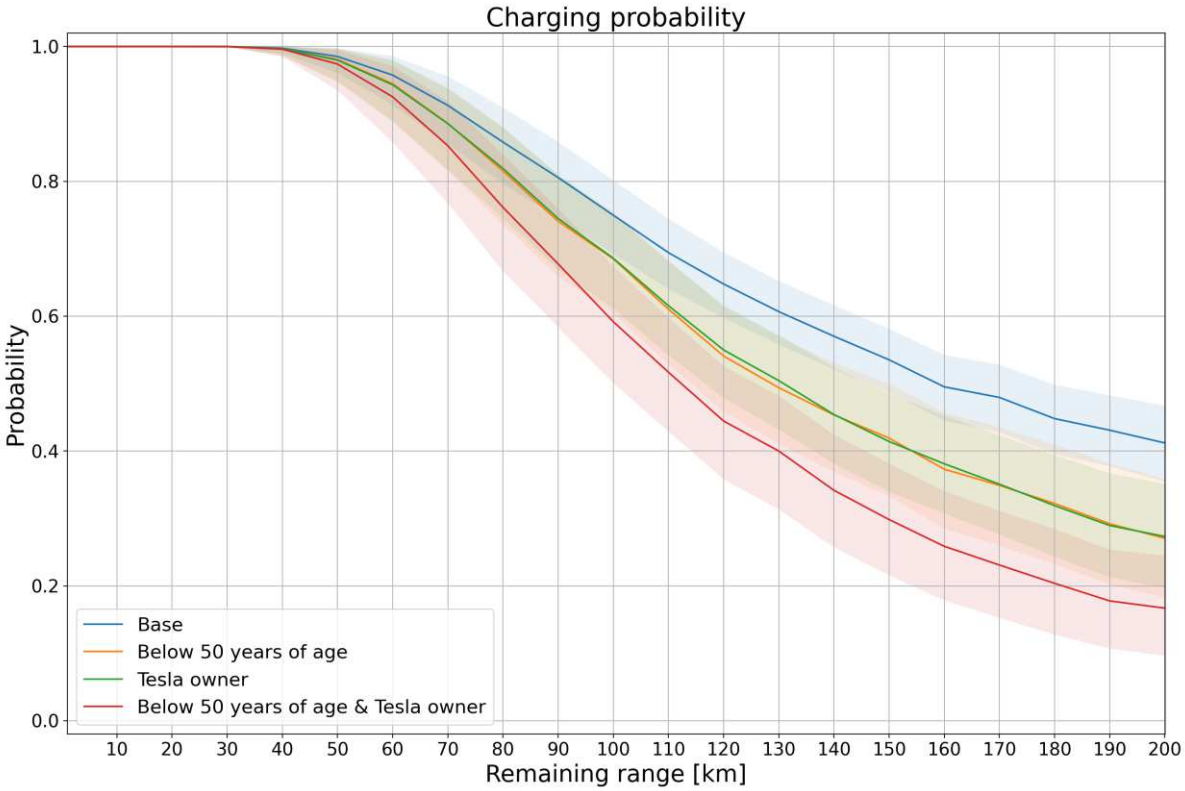


Figure 2: Probability to charge as a function of remaining battery range for various segments in the sample. Note that detour and waiting time are assumed to be zero.

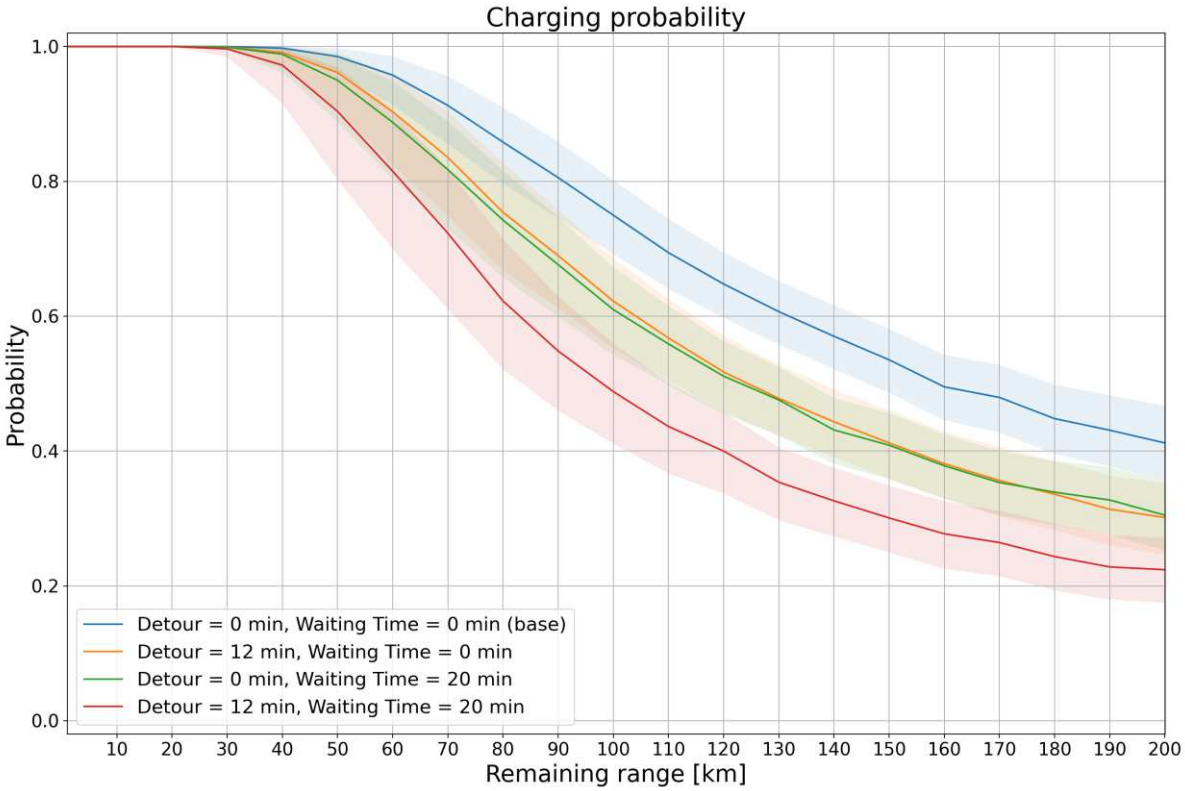


Figure 3: Average charging probabilities as a function of remaining battery range for various combinations of detour and waiting time.

We hold the belief that the Tesla segment and the youngest proportion of the sample, in many respects, represents first-mover segments in the EV market. This observation implies that over time, the preferences of the general population may align more closely with those of the first movers, leading to decreased emphasis on range anxiety concerns. However, range anxiety is strongly heterogenous over the population, and we should continue to see significant differences in the charging pattern of EV owners.

5 Conclusion

In the paper, we present an adaptive stated choice design for revealing charging choice behaviour in a long-distance travel situation. Users are presented with a choice set where new charging alternatives are dynamically introduced as the respondent moves forward while other alternatives are removed as they are 'passed'. This allows us to study charging behaviour and range anxiety jointly in a controlled experiment. Based on a mixed logit model, several findings are presented. We estimate a relationship between the probability of charging and the remaining battery range. The curvature of this relationship in this context where we consider a long-distance trip, presents itself as being different compared to the charging pattern of a typical day. It suggests that people are having a more strategic mind-set when deciding to charge in this context. On the one hand, many people are running their batteries closer to the limit before charging, while at the same time, a significant share of the sample charges at early stages when the battery is more than half full. As a result, the curve is less S-shaped when compared to everyday charging curves. We believe this is very realistic and complies with guidance offered in many commercial charging apps. It is the result of trade-offs along the route as regards waiting time and detours and the decision to charge.

Another conclusion is that range anxiety, and thereby battery utilisation, is indeed a heterogeneous quantity that will vary considerably between user groups as e.g., Tesla drivers are less prone to range anxiety compared to other drivers. If these represent first-movers in the EV market (Long et al., 2017; Hardman and Tal, 2016) and have charging preferences, which by time will be adopted by other

segments, it could suggest that range anxiety will become less as users become more familiar with EV technology and as the technology continues to advance. As a result, over time, the general population's preferences may align more closely with those of the first movers, reducing range anxiety concerns.

Finally, the paper taps into the ongoing debate of charging infrastructure design, which is very much a function of the user charging curves. If this curve differs from the conventional wisdom, it could potentially influence the placement of charging stations along the motorways.

5.1 Future research

The paper points to several avenues for future research within the area of EV charging, EV infrastructure design, and range anxiety.

First and foremost, with the data presented here having a limited sample size and presumably, in many ways representing a skewed population with many older male persons, it is of interest to redo this exercise for a different sample. Also, while in our case, we were not successful in representing a more complex choice design, in which price attributes were present, it should not obstruct others to go down this path. Although our findings could hint that prices are less of a concern for rare long-distance trips, it could also be that our design was simply not adequate to identify such effects.

Further study of heterogeneity in charging preferences should be highly appraised as it could have an impact on charging infrastructure design. While it may not be clear how, the impact can be studied in simulation models by using different charging curves.

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Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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Appendix A: Examples of stated-preference screens for the basic and complex design.

You now have **245 kilometers** remaining. You can choose between options B, C, D or not to charge right now.

You have between **145 - 165** kilometers left on your battery. What do you choose?

	B	C	D	Charge later
Distance to charching station	25 km	35 km	85 km	
Expected waiting time	5 min.	15 min.	20 min.	
Detour	9 min.	9 min.	0 min.	
Cost	6 kr./kWh.	6 kr./kWh	4 kr./kWh	
Charging speed	120 km/10 min.	90 km/10 min.	60 km/10 min.	
Facilities	Restroom & Restaurant	Restroom	Restroom & Restaurant	

You now have **215 kilometers** remaining. You can choose between options C, D, E or not to charge right now.

You have between **115 - 135** kilometers left on your battery. What do you choose?

	C	D	E	Charge later
Distance to charching station	5 km	55 km	85 km	
Expected waiting time	15 min.	20 min.	15 min.	
Detour	9 min.	0 min.	0 min.	
Cost	6 kr./kWh	4 kr./kWh	4 kr./kWh	
Charging speed	90 km/10 min.	60 km/10 min.	120 km/10 min.	
Facilities	Restroom	Restroom & Restaurant	Restroom	

You now have **185 kilometers** remaining. You can choose between options D, E, F.

You have between **85 - 105** kilometers left on your battery. What do you choose?

	D	E	F
Distance to charching station	10 km	55 km	75 km
Expected waiting time	0 min.	15 min.	15 min.
Detour	12 min.	0 min.	6 min.
Cost	4 kr./kWh	4 kr./kWh	6 kr./kWh
Charging speed	75 km/10 min.	120 km/10 min.	75 km/10 min.
Facilities	Restroom & Shop	Restroom	Restroom & Shop